



Health-System Specialty Pharmacy's Impact on Self-Administration of Denosumab During COVID-19

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Background

- According to the CDC, osteoporosis affects 1 in 4 women aged 65 and over and about 1 in 20 men aged 65 and over.¹
- Denosumab is a subcutaneous injection for osteoporosis. Due to the risk of adverse effects, denosumab must be administered by a healthcare provider once every six months.
- In March 2020, the World Health Organization declared the COVID-19 a global pandemic.² In response to this declaration, the manufacturer of denosumab, Amgen, temporarily warranted self-administration of the medication.
- To support treatment adherence, University of Chicago Medicine (UCM) Specialty Pharmacy aimed to safely transition patients from in-clinic to at-home administration, thus reducing patient risk for contracting COVID-19.
- The purpose of this project was to highlight initiation rates and adverse events for patients who transitioned to self-administer denosumab in the home.

Objectives

- Primary objective:** Initiation rates for patients who self-administered denosumab in the home.
- Secondary objective:** Adverse effects reported after self-administration in the home.

Methods

- Single-center retrospective review
- Study Duration: March 1, 2020 - November 30, 2020
- Eligible patients for transition to self-administration in the home:
 - Previously received denosumab doses in Endocrinology clinic or deemed appropriate and safe per clinic team for first dose injection
 - No reported adverse effects with previous doses
 - Able and willing to self-inject in the home
- Initiation rates and safety data collected using Therigy[®] and Epic Hyperspace

Figure 1: Referral Workflow between Endocrinology Clinic and UCM Specialty Pharmacy



Results

Figure 2: Denosumab Self-Administration Rates Following Referral to UCM Specialty Pharmacy

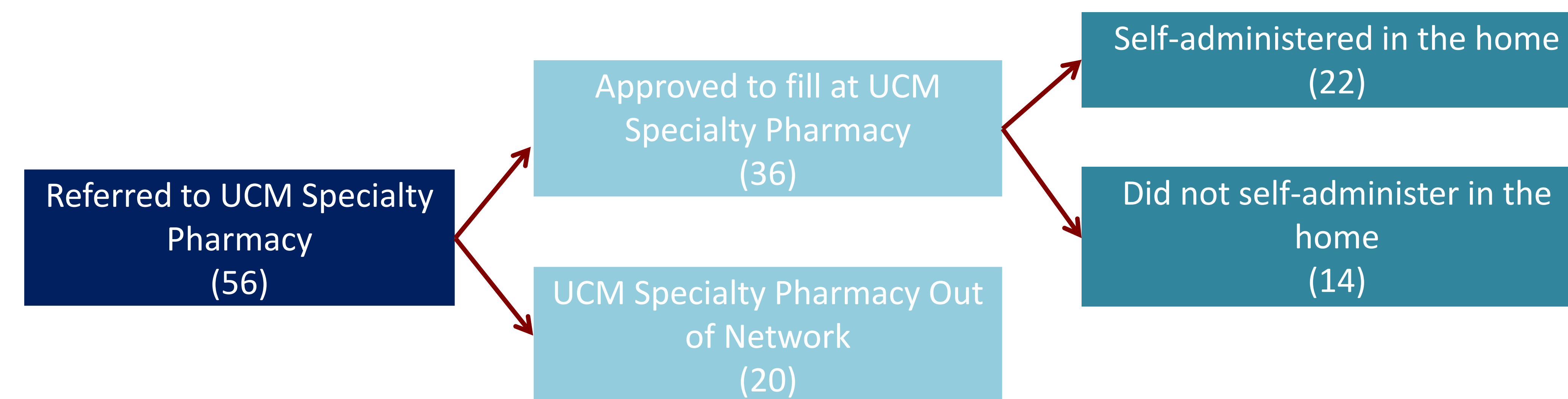
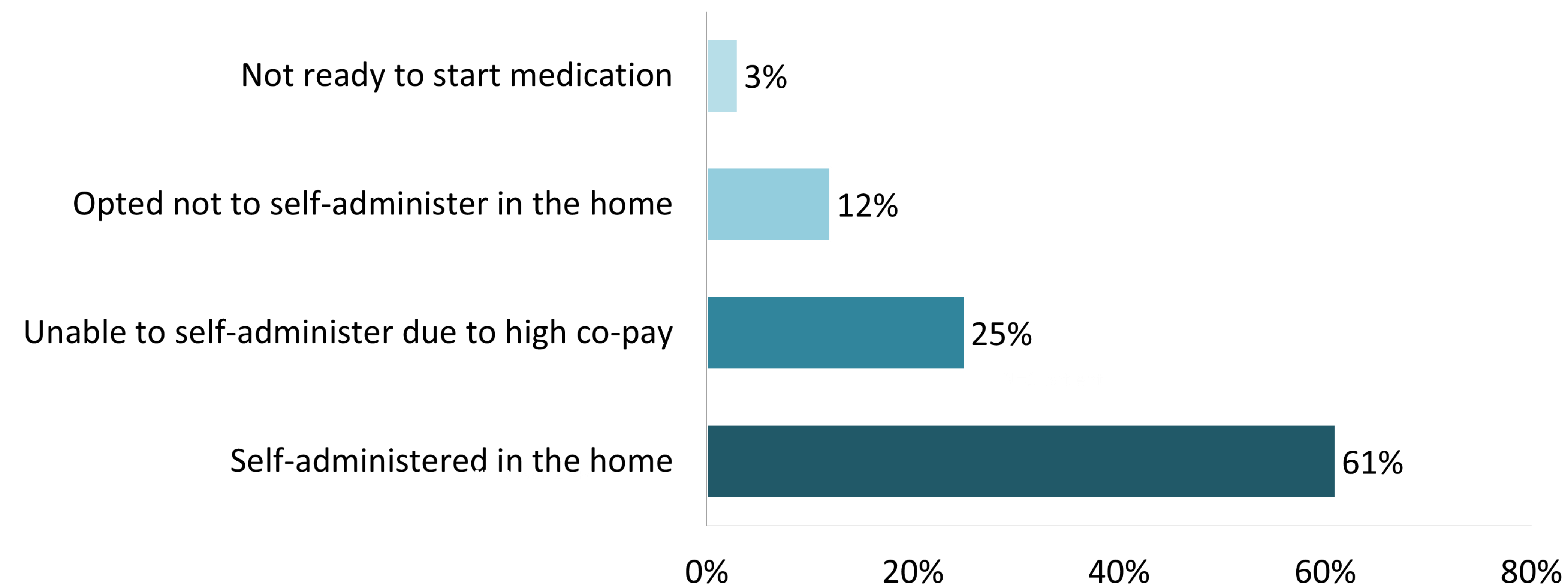


Figure 3: Denosumab Self-Administration Rates Following Approval to Fill at UCM Specialty Pharmacy (n=36)



36 patients were approved to fill through UCM Specialty Pharmacy

- 22 patients completed injection training and had denosumab delivered to their home for self-administration
- 14 patients did not qualify for self-administration in the home
 - 9 patients had a high co-pay, 4 patients opted not to self-administer and 1 patient was not ready to start medication

Table 1: Adverse Effects Reported after Denosumab Self-Administration in the Home

Outcome	Number of Patients
No Adverse Events, n (%)	11 (50)
Arthralgia, n (%)	7 (32)
Back Pain, n (%)	5 (23)

Zero patients experienced anaphylactic reactions, osteonecrosis of the jaw, hypocalcemia, atypical femur fractures, serious infections, or dermatologic reactions

Table 2: Turnaround Time for Patients who Self-Administered in the Home

Outcome	Days, Median (IQR)
Time to insurance approval	0
Time from referral placed to medication arriving at patient's home	5 (4-8)

Conclusions

- UCM Specialty Pharmacy was able to quickly and safely transition patients to self-administer denosumab in order to minimize the risks of potential COVID-19 exposure.
- Results show there were no serious adverse events in patients that self-administered denosumab in the home.
- The most common barrier to self-administration was a high co-pay.
- This review showcases the ability for a health-system specialty pharmacy to identify and execute a workflow that addressed a need within the community during a pandemic

References

- CDC –2021, March 3. Does Osteoporosis Run in Your Family? Retrieved from <https://www.cdc.gov/genomics/disease/osteoporosis.htm?CDCArefVal=https%3A%2F%2Fwww.cdc.gov%2Ffeatures%2Fosteoporosis%2Findex.html>
- World Health Organization (2021, January 29). Listings of WHO's response to COVID-19. <https://www.who.int/news/item/29-06-2020-covidtimeline>

Disclosures

The authors of this presentation have no financial interests with commercial entities that may have a direct or indirect interest in the subject matter of this presentation